FORMAL SMOOTHNESS DESCENDS ALONG A FAITHFULLY FLAT BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Preservation of formal smoothness by base change. Invoked at the start of the counted proof (line 38467-38468). Gives the easy direction of the equivalence.
- Characterisation of formal smoothness by a lifting condition on a two-term complex. Invoked twice in the counted proof (lines 38480-38500). Turns the descent question into exactness of a sequence after base change.

Source reference. Original lemma 00TJ: sources/stacks-algebra/pinned-original-source.tex, source lines 38037–38037. Navigation pointer for \ref{lemma-base-change-fs} used inside the counted span (line 38468); the argument that consumes it is retained in the span.

Source reference. Original lemma 031I: sources/stacks-algebra/pinned-original-source.tex, source lines 38128–38128. Navigation pointer for \ref{lemma-characterize-formally-smooth-again} used inside the counted span (line 38477); the argument that consumes it is retained in the span.

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Source reference. Original lemma 00RU: sources/stacks-algebra/pinned-original-source.tex, source lines 34320–34320. Navigation pointer for \ref{lemma-differential-seq} used inside the counted span (line 38486); the argument that consumes it is retained in the span.

Source reference. Original lemma 05A9: sources/stacks-algebra/pinned-original-source.tex, source lines 22679–22679. Navigation pointer for \ref{theorem-ffdescent-projectivity} used inside the counted span (line 38495); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $R \to R'$ be a faithfully flat ring map. Set $S' = S \otimes_R R'$. Then $R \to S$ is formally smooth if and only if $R' \to S'$ is formally smooth.*

Proof. If $R \to S$ is formally smooth, then $R' \to S'$ is formally smooth by Lemma 00TJ. To prove the converse, assume $R' \to S'$ is formally smooth. Note that $N \otimes_R R' = N \otimes_S S'$ for any S-module N. In particular $S \to S'$ is faithfully flat also. Choose a polynomial ring $P = R[\{x_i\}_{i \in I}]$ and a surjection of R-algebras $P \to S$ with kernel J. Note that $P' = P \otimes_R R'$ is a polynomial algebra over R'. Since $R \to R'$ is flat the kernel J' of the surjection $P' \to S'$ is $J \otimes_R R'$. Hence the split exact sequence (see Lemma 031I)

$$0 \to J'/(J')^2 \to \Omega_{P'/R'} \otimes_{P'} S' \to \Omega_{S'/R'} \to 0$$

is the base change via $S \to S'$ of the corresponding sequence

$$J/J^2 \to \Omega_{P/R} \otimes_P S \to \Omega_{S/R} \to 0$$

see Lemma 00RU. As $S \to S'$ is faithfully flat we conclude two things: (1) this sequence (without $'$) is exact too, and (2) $\Omega_{S/R}$ is a projective S-module. Namely, $\Omega_{S'/R'}$ is projective as a direct summand of the free module $\Omega_{P'/R'} \otimes_{P'} S'$ and $\Omega_{S/R} \otimes_S S' = \Omega_{S'/R'}$ by what we said above. Thus (2) follows by descent of projectivity through faithfully flat ring maps, see Theorem 05A9. Hence the sequence $0 \to J/J^2 \to \Omega_{P/R} \otimes_P S \to \Omega_{S/R} \to 0$ is exact also and we win by applying Lemma 031I once more. $\square$